

Advanced Calculus HW 6

Robert O'Connell

11th March 2026

Question 1

$$f(x, y, z) = (x^2 + y^2 + z^2)^\alpha$$

Partial Derivatives at $(0, 0, 0)$

$$\begin{aligned} f_x(0, 0, 0) &= \lim_{h \rightarrow 0} \frac{f(0 + h, 0, 0) - f(0, 0, 0)}{h} \\ &= \lim_{h \rightarrow 0} \frac{h^{2\alpha} - 0^\alpha}{h} \\ &= \lim_{h \rightarrow 0} h^{2\alpha-1} - \lim_{h \rightarrow 0} \frac{0^\alpha}{h} \end{aligned}$$

This limit only exists if $\alpha > \frac{1}{2}$

The same is true for y and z , (nothing changes for y and z because of the question)

$$\lim_{(\Delta x, \Delta y, \Delta z) \rightarrow (0, 0, 0)} \frac{\Delta f - f_x(0, 0, 0)\Delta x - f_y(0, 0, 0)\Delta y - f_z(0, 0, 0)\Delta z}{\sqrt{\Delta x^2 + \Delta y^2 + \Delta z^2}}$$

with

$$\Delta f = f(\Delta x, \Delta y, \Delta z) - f(0, 0, 0) = (\Delta x^2 + \Delta y^2 + \Delta z^2)^\alpha \text{ for } \alpha \neq 0$$

With $\alpha > \frac{1}{2}$, we then have the partial derivatives as

$$f_x(x, y, z) = 2x\alpha(x^2 + y^2 + z^2)^{\alpha-1}$$

$$f_y(x, y, z) = 2y\alpha(x^2 + y^2 + z^2)^{\alpha-1}$$

$$f_z(x, y, z) = 2z\alpha(x^2 + y^2 + z^2)^{\alpha-1}$$

Then

$$f_x(0, 0, 0) = f_y(0, 0, 0) = f_z(0, 0, 0) = 0$$

Then with $\alpha > \frac{1}{2}$ the earlier limit is defined and is

$$\begin{aligned} &\lim_{(\Delta x, \Delta y, \Delta z) \rightarrow (0, 0, 0)} \frac{(\Delta x^2 + \Delta y^2 + \Delta z^2)^\alpha}{(\Delta x^2 + \Delta y^2 + \Delta z^2)^{\frac{1}{2}}} \\ &= \lim_{(\Delta x, \Delta y, \Delta z) \rightarrow (0, 0, 0)} (\Delta x^2 + \Delta y^2 + \Delta z^2)^{\alpha - \frac{1}{2}} \end{aligned}$$

Which is equal to zero $\forall \alpha > \frac{1}{2}$
Thus f is differentiable $\forall \alpha > \frac{1}{2}$

Question 2

$$L(x, y, z) = f(x_0, y_0, z_0) + f_x(x_0, y_0, z_0)(x - x_0) + f_y(x_0, y_0, z_0)(y - y_0) + f_z(x_0, y_0, z_0)(z - z_0)$$

Then,

$$\begin{aligned} L(x_0, y_0, z_0) &= f(x_0, y_0, z_0) \\ \Rightarrow f(0, -1, -2) &= L(0, -1, -2) = -4 \end{aligned}$$

Also,

$$\begin{aligned} L_x(x, y, z) &= f_x(x_0, y_0, z_0) \\ \Rightarrow f_x(0, -1, -2) &= \frac{\partial}{\partial x}(x + 2y + 3z + 4) = 1 \end{aligned}$$

Similarly,

$$\begin{aligned} f_y(0, -1, -2) &= 2 \\ f_z(0, -1, -2) &= 3 \end{aligned}$$

Question 3

(a)

$$z = V(r, h) = \pi r^2 h$$

Then

$$\begin{aligned} dz &= f_r dr + f_h dh = 2\pi r h dr + \pi r^2 dh \\ \Rightarrow |dz| &\leq |2\pi r h| |dr| + |\pi r^2| |dh| \end{aligned}$$

Then for a cylinder with $r = 5$, $h = 3$ and $dr = 0.1$ and $dh = 0.05$

$$|dz| \leq |30\pi| |0.1| + |25\pi| |0.05| = 4.25\pi$$

(b)

With $z = f(x_1, x_2, \dots, x_n)$

We have

$$dz = \sum_{i=1}^n \frac{\partial f}{\partial x_i} dx_i$$

Then,

$$\begin{aligned} |dz| &= \left| \sum_{i=1}^n \frac{\partial f}{\partial x_i} dx_i \right| \\ &\leq \sum_{i=1}^n \left| \frac{\partial f}{\partial x_i} \right| |dx_i| \end{aligned}$$